

Isaac Mubiru

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Kampala, Uganda (Remote, UK overlap UTC+3)

PROFESSIONAL SUMMARY

Full-stack engineer (Node.js + TypeScript, React + TypeScript) with 6+ years turning messy operational workflows into automated production systems. Independently delivered complex systems end to end: led the zero-downtime migration of 20+ legacy applications to Firebase/GCP reporting directly to the CTO, built a USSD legal-aid service for low-tech field operators, and integrated OpenAI + Google Vision to automate prescription intake over WhatsApp. High-autonomy remote operator — hired as a solo contractor, converted to a CTO-direct role. Now applying AI-driven workflow automation to property operations.

CORE COMPETENCIES

Node.js + TypeScript

React + TypeScript

AI / LLM Integration (OpenAI)

Workflow Automation

End-to-End Ownership

Legacy System Migration

SQL & Data Modeling

GCP · Firebase · Docker · Kubernetes

WORK EXPERIENCE

- MTailor

Sep 2022 – Present

Software Engineer — US Remote

 - Independently led the end-to-end migration of 20+ legacy applications from Parse/MongoDB to Firebase/GCP with zero downtime, reporting directly to the CTO — untangling fragmented legacy workflows while the business kept running
 - Implemented real-time two-way sync between MongoDB and Firestore using Node.js and Google Pub/Sub, letting old and new systems run in parallel during cutover
 - Saved \$5,000/month by migrating all services off AWS; automated file-storage migration from S3 to GCS with Python
 - Implemented Express Shipping end to end, generating an additional \$40 revenue per order
 - Built a 3D visualisation feature using video overlay and ffmpeg, increasing buyer conversion
- MTailor

Mar 2022 – Jul 2022

Backend Contractor — US Remote

 - Hired via Upwork to independently execute a complex backend migration; delivered it, documented the new Firebase SDKs for other engineers, and trained the Ops team — then converted to full-time
- Mind2matter

Jan 2022 – Jul 2022

Software Engineer — US Remote

 - Built Node.js backends for DeFi applications and delivered pixel-accurate React UIs from Figma under tight agency timelines
- Dr Wealth

Aug 2021 – Jan 2022

Software Engineer — Singapore Remote (FinTech)

 - Kept real-time stock prices current across 2M+ Firestore records via automated jobs against Morningstar APIs; extended APIs with Cloud Functions and Express.js; shipped responsive React + Tailwind pages
- CodeBits

Jan 2020 – Jul 2021

Software Engineer & Team Lead — Uganda

 - Automated deeply manual, operationally complex workflows for NGOs: case management for FIDA Uganda and a paralegal database for LASPNET, leading a team of 4 developers
 - Built a USSD service so legal-aid providers on feature phones could reach a Node.js backend via gRPC — technology bent to fit real field operations
 - Architected a microservices backend with Apache Kafka, Docker, and Kubernetes; shipped a React Native mobile app with push notifications

PROJECTS

Healthcare WhatsApp Chatbot — AI Workflow Automation

Integrated OpenAI and Google Vision for prescription OCR via the WhatsApp Business API — replaced a fully manual intake workflow with an automated conversational pipeline.

Node.js · OpenAI API · Google Vision · WhatsApp Business API

Real-Time MongoDB ↔ Firestore Sync Pipeline

Live two-way sync supporting parallel production traffic during a zero-downtime replatform of 20+ apps.

Node.js · Google Pub/Sub · MongoDB · Firestore

Marketplace Recommendation Engine

Collaborative + content-based filtering engine in MongoDB for service-provider matching.

Node.js · MongoDB

EDUCATION

BSc Computer Engineering — Makerere University

2018 – 2022

Kampala, Uganda

SKILLS

Proficient: Node.js, TypeScript/JavaScript, React, Firebase, GCP, SQL, MongoDB, Python, Git, Linux

Intermediate: Go, AWS, Rust, React Native **Languages:** English, Kiswahili